Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/25/2019
PI Name:	Zawieja, David Ph.D.		
Project Title:	Effects of Microgravity on Lymphatic Structure and Function		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date corrected to 9/08/2017 per F. Hernandez/ARC (Ed., 6/26/17) NOTE: End date extended to 8/09/2017 per NSSC information (Ed., 3/10/17) NOTE: End date changed to 3/8/2017 (original end date was 3/8/2016) per NSSC information (Ed., 3/17/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cromer, Walter Ph.D. (Texas A&M University System)		
Grant/Contract No.:	NNX15AG54G		
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Performance Goal Text:			

Task Description:	This project combines aspects of two proposals that were selected for funding: “Effects of microgravity on lymphatic proliferation and transport efficiency in the gastrointestinal system of C57/BL6 mice” (#14-14SF_Step2-0079) and “Tissue Sharing (B7): Effects of Microgravity Adaptations on Cephalic Lymphatic Function and Associated Edema Development and Immune Dysfunction” (#14-14SF_Step2-0092). Both of these proposals have at their basis the idea that long-term space flight inhibits lymphatic function, which will impair the transport of immune cells, fluids, and macromolecules from the interstitium. This lymphatic inhibition could affect many aspects of astronaut health, especially upper body edema, CSF (cerebralspinal fluid) pressure regulation, visual acuity loss, gastrointestinal dysfunction/inflammation, and immune impairment due to diminished transport of fluid, large molecules, and immune cell transport to and from the nodes. The visual acuity disturbances from space flight are thought to be related to the increases in intracranial pressure because of the space flight induced cephalic fluid shift. Cephalic lymphatic function has been intimately linked to the regulation of cerebrospinal fluid (CSF) pressure via the lymphatic removal of CSF through the cribriform plate. The cribriform plate and associated lymphatic network moves lymph derived from CSF towards the cervical lymph nodes en route to the veins of the neck. This provides part of a multi-faceted CSF pressure regulatory mechanism as well as a modulator of neurological immunity. We will study the role of the cephalic lymphatic network in CSF pressure dysfunction and the assessment of correlative immunological components. Space flight is also known to cause alterations in the development and function of the gastrointestinal (GI) tract. This is associated with lymphostasis, gut edema, reduced fat deposition in the mesentery, and visceral organs, and reduced size of immune associated organs, similar to what is seen in intestinal lymphangiectasia. All of these symptoms are also found in patients with GI inflammation. Intestinal inflammation has not been studied in human subjects during space flight, but we have seen that simulated space flight resulted in the development of intestinal inflammation in animal models. However, this has not been measured in the same space-flown animal model. The role of GI lymphatic dysfunction, which is likely a causative factor in many of these pathologies has NEVER been systematically assessed. Thus we propose that space flight induced lymphatic dysfunction is responsible for the development of edema and inflammation in multiple tissues, specifically producing dysfunction in CSF pressure regulation and GI function. The previous applications both proposed to test this general hypothesis using very similar experimental protocols applied to the head/neck and GI tissues respectively for #14-14SF_Step2-0092 and #14-14SF_Step2-0079. In the combined grant we propose 3 modified Aims to test this general hypothesis: Aim 1--To analyze changes in the structure and functional implications of the pre and postnodal cephalic lymphatic architecture due to space flight-induced adaptations.(old Aim 1 of #14-14SF_Step2-0092); Aim 2--To evaluate space flight adaptations of the network structure of lymphatic vessels of the cribriform plexus and the GI tract. Increases in lymphangiogenesis are commonly associated with inflammation and lymph transport insufficiency. We will determine alterations in lymph vessel number, size, and distribution in these tissues. We will also determine if there are alterations in the formation of lymphatic valves, which are an important structural component of lymphatic vessels. (old Aim2 of #14-14SF_Step2-0092 combined with old Aim 1 of #14-14SF_Step2-0079); Aim 3--To analyze changes in lymph node structure that could alter the delivery of important immunological components to and from the nodes (cephalic for the head and neck tissue and mesenteric node for the GI tissue) leading to immune dysregulation and modulation of nodal lymph flow. We will also determine if there are shifts in the cell populations that reside in the lymph nodes, and around the prenodal lymph vessels. (old Aim 3 of #14-14SF_Step2-0092 combined with Aim 2 of #14-14SF_Step2-0079).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will provide information on the impact of altered lymphatic function and how this may affect different tissues. Specifically we will be studying this impact on GI tissue and the development of gut inflammation, CSF pressure regulation and associated visual acuity losses, and lymph transport of immune signals to and from the nodes and its implication to immunity. These issues are associated with problems seen with astronaut exposure to the space environment as well as other Earth bound pathologies.
	FINAL REPORT SUMMARY (April 2019--Compiled from investigator's 2017 final report) We were not able to generate the type of histological data that we wished to obtain from this tissue; however, we were able to perform measures that do give us an insight into the fluid balance and tissue inflammatory state of the GI system in longer term rodent space flight. Our histological analysis showed that there was immune cell infiltration into the flight colons to a greater extent than in ground control animals. This suggests an inflammatory reaction occurring within the tissue in flight. Likewise, the colon structure (length to weight ratio) also supports the development of an inflammatory response resulting in edema. This is supported by both histological analysis and wet to dry weight ratio of the colon. There were similar gross signs of tissue changes in the ileum but to a lesser extent. There was also no way to compare the immune cell infiltration in the ileum to the colon due to even greater disruption of tissue structure than in the colon. Our observations on the types of immune cells invading the colon were hampered by the disruption the tissue handling caused to the tissue and antigens in the tissue. We observed many cells with large nuclei and small cytoplasmic volumes entering the tissue of space flown animals at plexi located in the submucosa and between the longitudinal and circumferential muscle layers. Due to the morphology of these cells we hypothesize that these cells are a mixture of monocyte derived cells and t-cells. There were few to no trilobed or barbell shaped nuclei that would be indicative of neutrophil or basophil/eosinophil infiltration. There were a number of granuloma like structures, which penetrated the submucosa into the mucosa and in extreme cases the lumen of the bowel in both flight and baseline animals. The only comparable structures that we found in ground control animals appeared to be expansions of normal lymphoid tissue. The changes in the histidine levels in the stool and tissue was an analysis done based upon a hypothesis that the changes reported in the microbiome of space flown rodents would reduce the availability of this critical metabolite. Histidine, derived from the microbiome is converted to histamine in the tissue and opposed to allergic reactions in other tissues actually serves to regulate immune responses in the gut. Additionally, histamine is critical to lymphatic vessel dilation which allows the vessels to have lower resistance to transporting fluid from the tissue. When we couple the gut cytokine profiles (predominantly lower cytokine patterns) together with the rest of the data that indicate some gut inflammation, this appears to be counterintuitive. However, the levels of cytokine (inflammatory or otherwise) in any tissue depend on the rates of delivery and generation of those cytokines to/in that tissue to the rate of removal of those cytokines (by degradation and lymph removal) from that tissue. When you couple the likely enhanced production of inflammatory cytokines in the gut with the lower levels found, one potential explanation is that lymph removal from the gut tissue is elevated during flight with the loss of the effects of gravity on lymph flow. Unfortunately, the histology of the head/neck

Task Progress:

tissues was of even less use in terms of inflammatory scoring, immunohistochemistry, and flow cytometry.

Meanwhile the pattern of the effects of flight on the inflammatory cytokine profiles showed an elevation in their levels in the cribiform plate and cervical lymph node. That is just the opposite of what was seen in the gut cytokine levels during flight. Explanations to this include possible increased production or delivery of the cytokines to the local tissues or an impaired removal from these tissue via lymph flow. This fits data we have previously seen in ground based models of microgravity in terms of impaired lymph flow generation predominantly in the head/neck region. Whether this is the explanation of these data remains to be seen with better measures of lymph flow and lymph flow generation during flight.

In summary, we have observed signs of inflammation and fluid imbalance in the tissue of space flown RR3 animals and from biochemical analysis there appears to be an imbalance of microbiome produced metabolites in the gut and changes in tissue cytokine profiles that are opposite in the head/neck versus the gut. These changes are likely linked and may feed into one another but to test this hypothesis requires more studies. First, we would need to confirm that there are changes in lymph transport in the gut and head/neck; second, we would have to test if other local measures of tissue inflammation match the cytokine changes we observed. Likewise, we need to test if the microbiome in and of itself can induce this inflammation/lymphatic disruption, and finally we would have to test if the restoration of either the microbiome or constituent metabolic products resolves these issues.

ANNUAL REPORT--RR3 progress report 2/28/2017

This project was originally delayed because of problems with flight and astronaut time availability, so we asked for the first no-cost extension (NCE) a little over a year ago since we had no access to flight tissues. Since then we began to start work on this project with access to tissues from flight animals (RR3) in September 2016 . Given the associated tissue storage approaches used for the RR3 flight animals, we had to implement changes to our analyses to accommodate the changes in tissue acquisition protocols associated with the new flight. We replicated the freeze thaw process that was used on the tissue that we will receive access to and began to implement changes to our originally described protocols and/or developed new analyses to accomplishes most of the goals of our original approved studies. We have participated in the meetings that lead to the Tissue Sharing opportunity in RR3 and completed the tissue acquisition at University of Colorado Boulder. We are currently finishing the use of these tissues and, as we thought, had to significantly modify and develop new analytical approaches. Our modified RR3 project called for analysis of lymphatic and associated tissues in 2 body regions: 1. the cervical lymphatics, cribriform plate tissues, meningeal tissues, and cervical lymph nodes (associated with potential changes in CSF pressure and visual acuity issues) ; 2. The colon small intestine, mesentery, and mesenteric lymph node (associated with potential changes in inflammatory status of the gut).

There were 4 general analytical approaches to these tissues: 1.Gross tissue measures (weight, length, size, etc.) ; 2.Classic histological evaluation of tissue inflammatory status (i.e., tissue inflammatory scores) ; 3. Immunofluorescent analyses of different markers indicating various cell and inflammatory indicators ; and 4. Various biochemical analyses of the tissues inflammatory status via Elisa and cytokine analyses.

The biggest concern we had was the potential damage cause by the freeze methods employed in the RR3 mission. This was a particular concern for the histology and immunofluorescence analysis.

We have completed a number of physical measurements that we have taken from the tissues that provide a insight into the status of the tissue. We have the weights of all the relevant bowel tissues (small bowel, large bowel, caecum, and mesenteric lymph node). We have also calculated the volume of the mesenteric lymph nodes of the animals, which has allowed us to determine the density of the nodes as well. We have calculated the percent water weights of the large bowel tissue to measure edema and we have also measured the distance between nuclei of the muscularis mucosa (only portion of the bowel wall with little to no damage) of the bowel wall as an objective histological method to determine tissue swelling.

To date, our assessment of histological structure and immunofluorescent antigen staining ability has been completed for colon and cervical lymph node tissues; we are about done with the small intestine and mesenteric lymph node histology. The original application for the GI tissues called for 3 general measures-- 1) vascular structure, 2) immune cell number and status, 3) physical and histological changes. Due to the damage of the tissue due to freezing and handling (any gross movements, bending, straightening, etc., of the gut after freezing leads to accumulating damage) before we were allowed to excise the samples that we received several of these measures were impossible to obtain. Histological structural integrity of the colon and small intestine has shown significant damage due to the freezing process to the intestinal epithelium and lamina propria regions (when comparing to similarly analyzed animals, minus the freezing process). These regions are critical for intestinal barrier function, but also are important for assessing the extent of both vascular structures and immunological state. So any assessment of any impact of space flight will be heavily skewed due to the tissue preservation. Thus we have lost the ability to score more than half of the GI histological based markers of tissue damage for an inflammatory score of the GI tissues. We will not be able to accurately provide full histological data as the epithelial and most of the crypt structures were damaged; additionally, we could not provide data on lipid absorption due to the same factors and rupturing of the fine subcellular structures of villi and the nascent chylomicrons. We note that the histological freeze damage is also similar in the cribiform plate tissues. This appears less true for the lymph node structures; however, there are notable signs of poor tissue preservation/freezer damage that also may bias/skew the results. We are continuing the histology of the mesenteric nodes to look for gross changes. This analysis should be done in a few weeks.

15+ immunofluorescent markers for blood and lymphatic markers, vascular muscle markers, and immune cell markers have been characterized in RR3 tissues. Nearly all have succeeded in terms of positive staining; however, particular markers (such as LYVE1) did not work in the RR3 tissues, and some markers (mostly immune cell markers) showed peculiar morphology of cellular staining that may be tied due to the freezing process. The loss of the lymphatic specific marker LYVE1 is particularly problematic as it is the principle markers used in the cerebral tissue lymphatic structural analysis. We are continuing to attempt to use some antigen retrieval steps to rescue this important lymphatic marker; however, to date those have been successful. The implications of the markers that did not work or show unusual staining will be considered as part of the final assessment of flight tissues.

Due to the same issues of damage and potential loss of antigenic epitope, the ability to provide accurate information on vascular structures in the bowel is questionable; however, we are examining other routes to provide some of this data currently (staining of the mesentery, etc.). Loss of immune cell antigenicity is also an issue and detection of these cells immunohistochemically may not be possible in some cases. We are utilizing different methods (differential histology

	staining, Western blot, nuclear counts, etc.) as alternate methods to provide relevant data. We are just beginning to conduct the biochemical assays and we are hoping to rely on some extra biochemical markers of inflammation from these tissues to try to offset some of the problems the freezing method had on the tissue histology-based assays. We will be assessing if the freeze methods employed on RR3 affect these assays using control animals frozen optimally as compared to those similar to those that have been employed on RR3. After completing some preliminary tests we have just ordered the assay kits for the analysis of tissue extracts from the large and small bowel and fecal samples (cytokine and calprotectin, maybe albumin and possibly IgM in stool). These data will provide valuable information on inflammatory status of the bowel tissue and supplement the above mentioned measures that we already have. In summary we still have another 4+ months of analysis to complete on the RR3 tissue samples, which is the rationale for the grant's second no cost extension.
Bibliography Type:	Description: (Last Updated: 04/24/2019)
Articles in Peer-reviewed Journals	Cromer WE, Zaweija DC. "Acute exposure to space flight results in evidence of reduced lymph transport, tissue fluid shifts, and immune alterations in the rat gastrointestinal system." Life Sci Space Res. 2018 May;17:74-82. https://doi.org/10.1016/j.lssr.2018.03.005 ; PubMed PMID: 29753416 , May-2018
Articles in Peer-reviewed Journals	Cromer WE, Zawieja SD, Doersch KM, Stagg H, Hunter F, Tharakan B, Childs E, Zawieja DC. "Burn injury-associated MHCII(+) immune cell accumulation around lymphatic vessels of the mesentery and increased lymphatic endothelial permeability are blocked by doxycycline treatment." Lymphat Res Biol. 2018 Feb;16(1):56-64. Epub 2018 Jan 23. https://doi.org/10.1089/lrb.2017.0032 ; PubMed PMID: 29359999; PubMed Central PMCID: PMC5810432 , Feb-2018